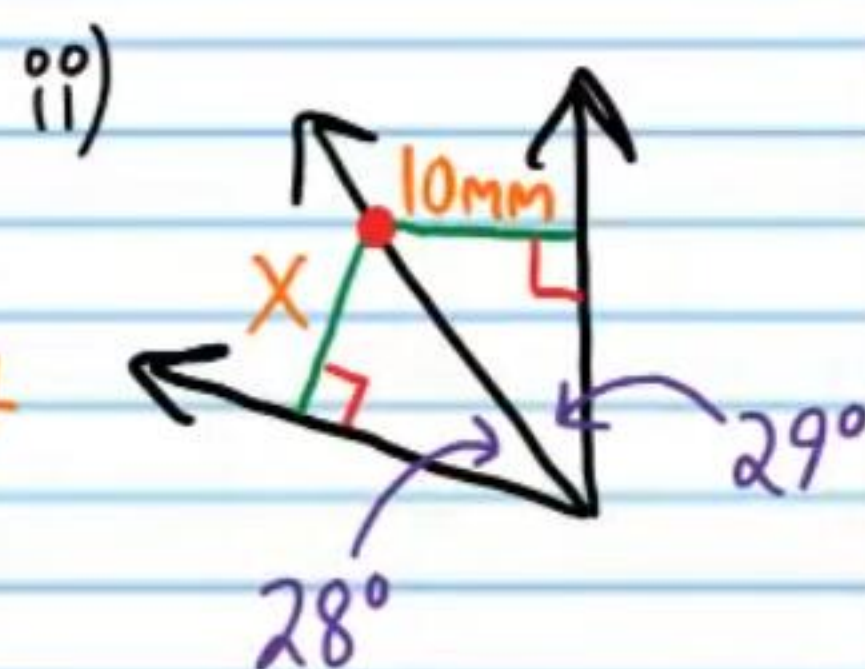
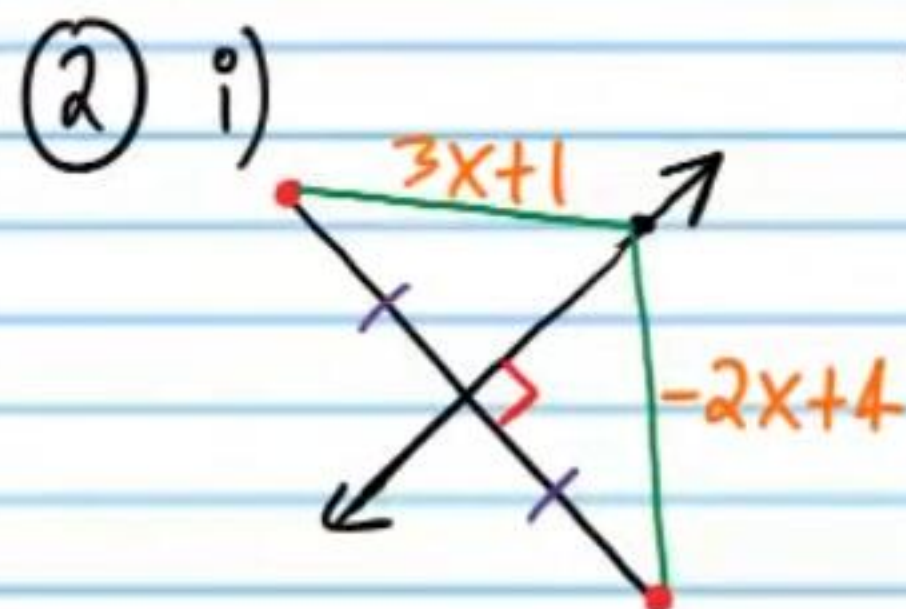
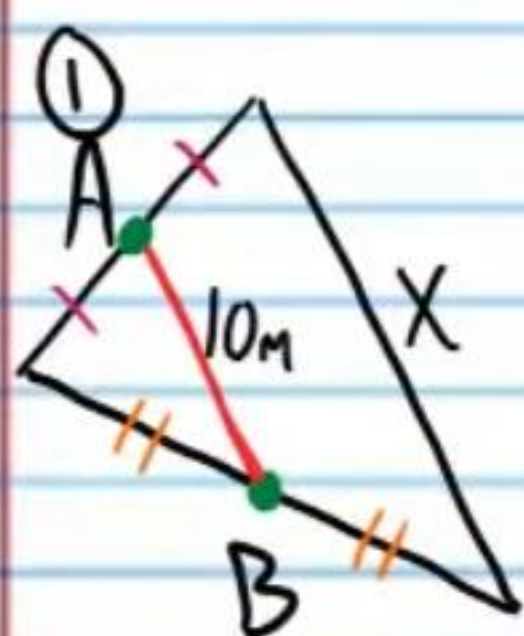


Practice Geometry Final Test-Part I (Version 2)

Name
Date
Course

- I) Write your name, the date, and the course title (i.e. Geometry).
- II) Show all work.
- III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) You may use a calculator for some problems on this test.
- V) Each problem is worth four points for a total of 100 points.

In problem #'s 1-2, find X .



- ③ Predict the next thing in the lists below.
- 3, -30, 300, -3000, ...

- ④ If $(3, 8)$ is the midpoint between points A and B, and the coordinates of B are $(-2, 1)$, find the coordinates of A.

- ⑤ Use the following information to make a conjecture.

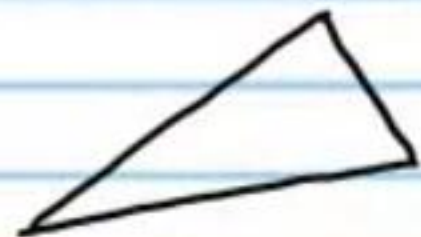
Walter gave Alice a gift for Valentine's Day last year and the year before.

- ⑥ i) What law of logic can be used to form a conclusion generated by the information below?

If Tom drinks some water, then he will feel better. If he feels better, then he will do some gardening in the remaining part of the day.

ii) What type of reasoning is used below?

By the definition of a triangle, we know that the polygon below is a triangle.



⑦ Write a counterexample for the statement below.

If Fido came back to the house soaking wet, then it must have rained.

⑧ Match a, b, or c for parts i-iii below.

Original: If I finish my homework, then I will go surfing.

* Remember these statements do not have to make sense.

i) If I go surfing, then I will finish my homework.

a) Converse

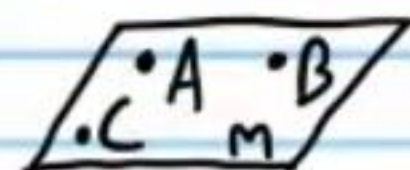
ii) If I do not go surfing, then I will not finish my homework.

b) Contrapositive

iii) If I do not finish my homework, then I will not go surfing.

c) Inverse

iv) What are two ways to name the plane below?

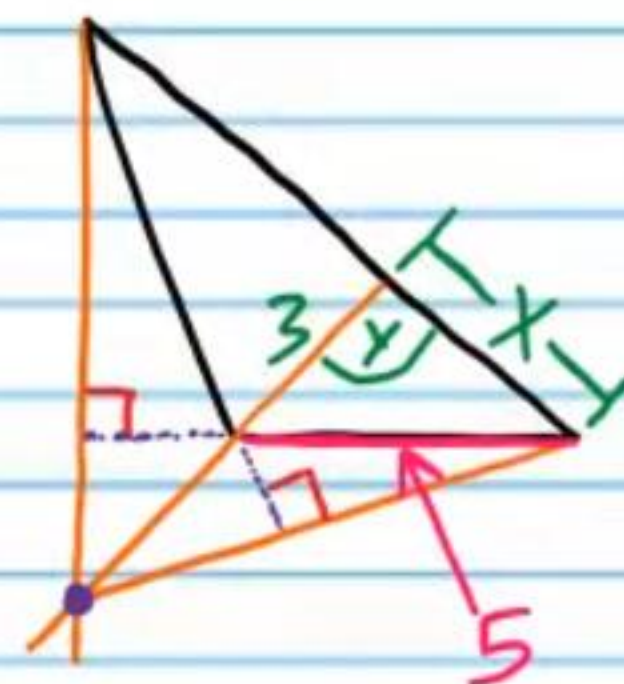


⑨ Given the definitions below, write $\sim q \rightarrow \sim P$ in word form.

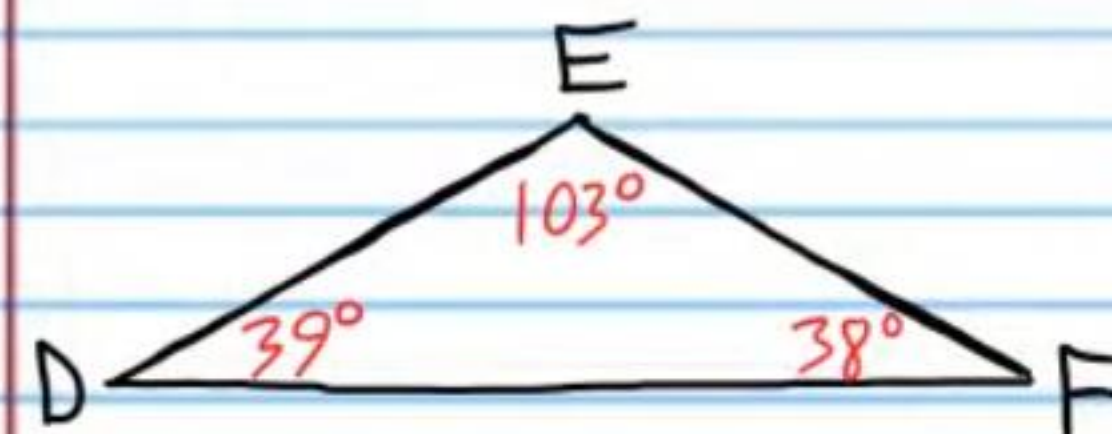
P: I will stay home.

q: It is not raining.

⑩ In the following problem, find X and y.



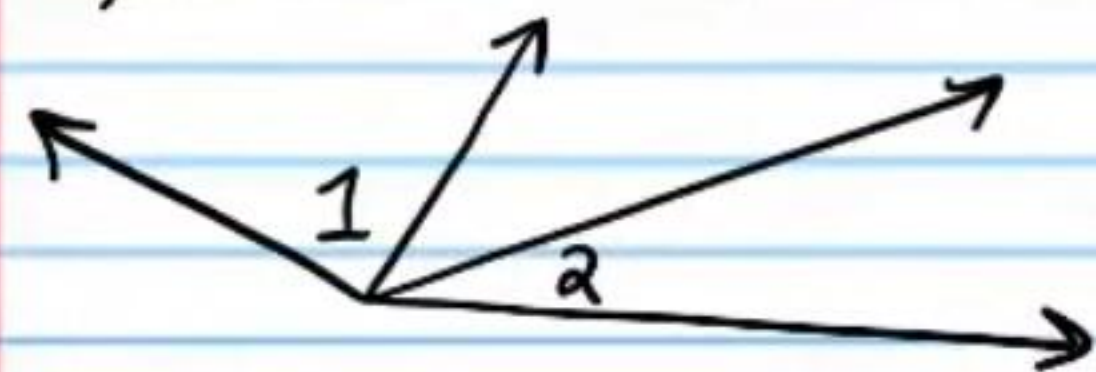
⑪ List the sides of the triangle below from shortest to longest.



12) i) Circle each figure below that is a polygon.



ii) Are $\angle 1$ and $\angle 2$ adjacent? Explain.



13) i) If the triangle below were classified by its sides, what word would be used to describe it?



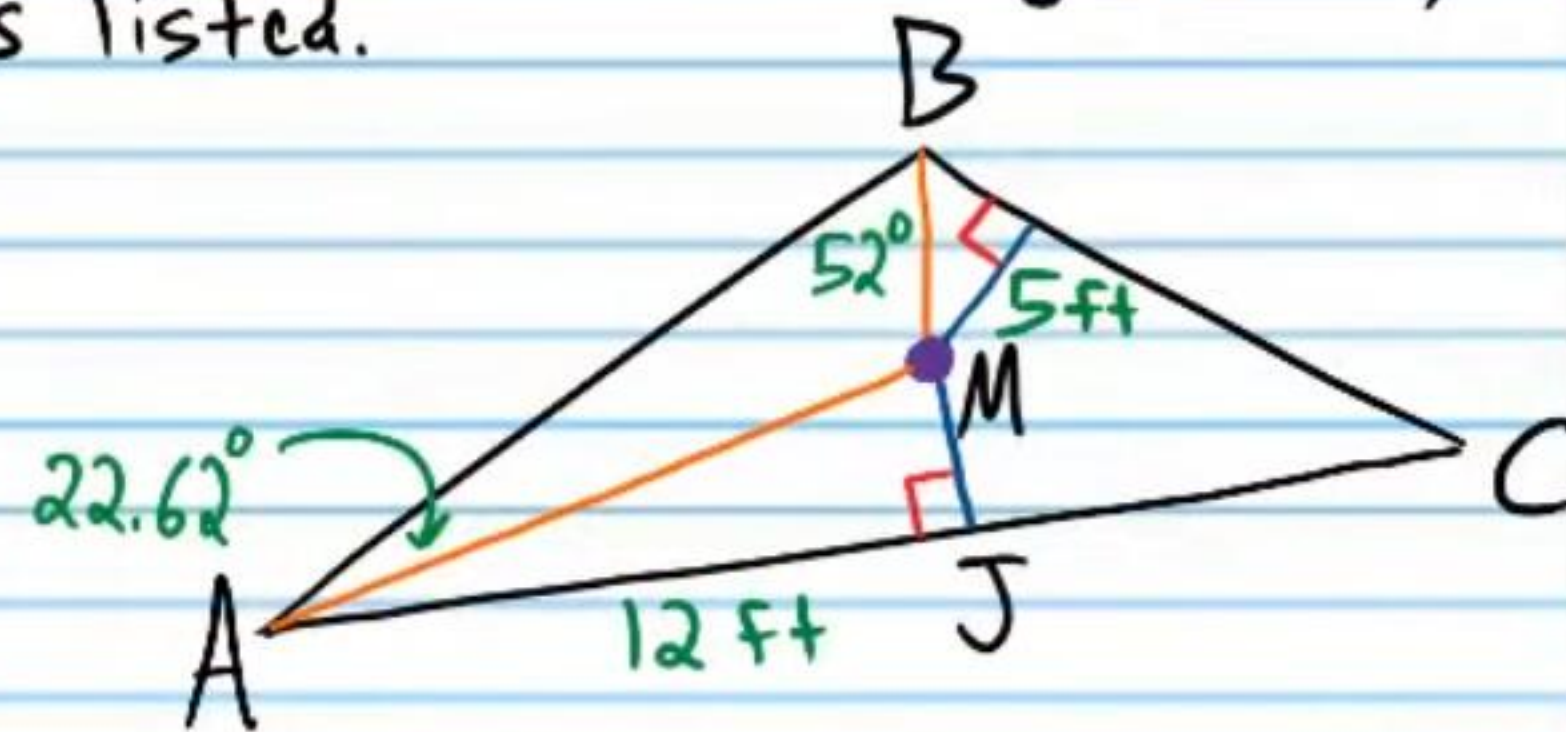
ii) If the triangle above were classified by its angles, what word would be used to describe it? There are two possible answers. Choose one.

14) If point M is the incenter of the triangle below, find the quantities listed.

a) $m\angle MBC$

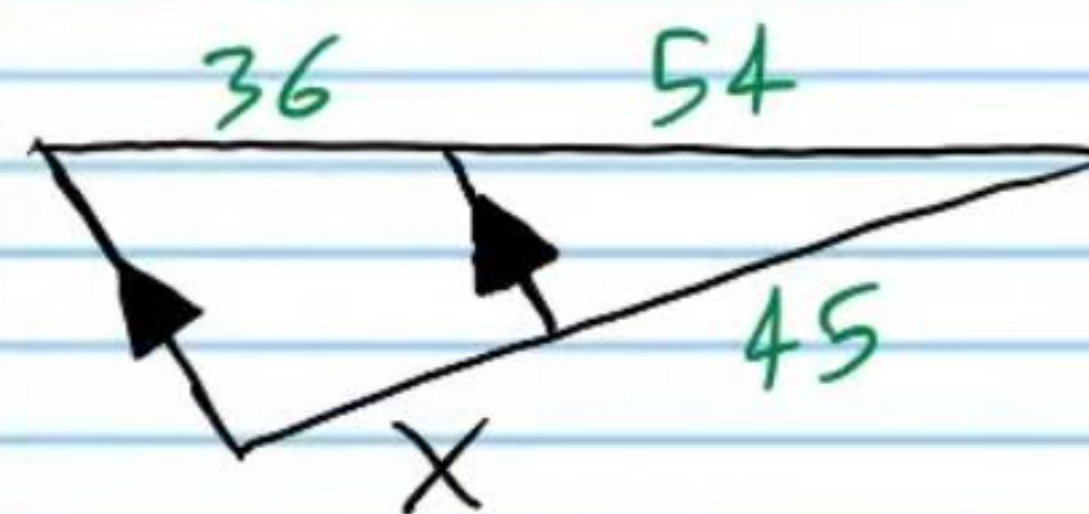
b) MJ

c) $m\angle C$



d) AM

15) Find X .

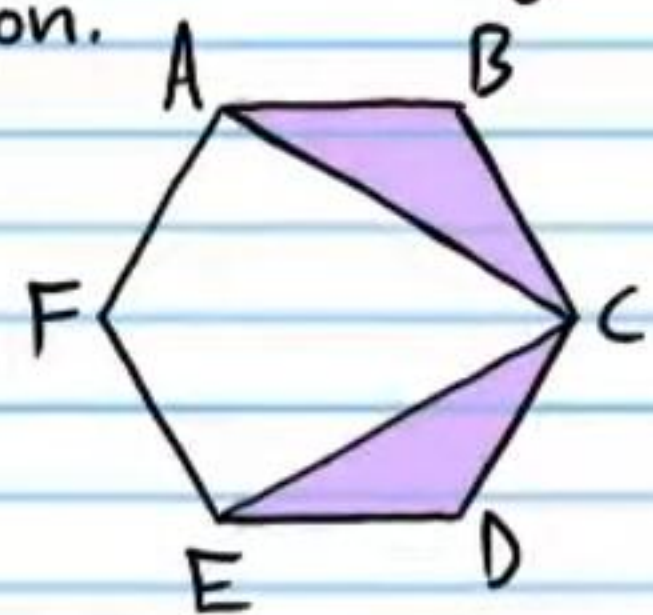


⑩ If the numbers below represent side lengths of a triangle, write an inequality that shows the possible lengths of the third side.

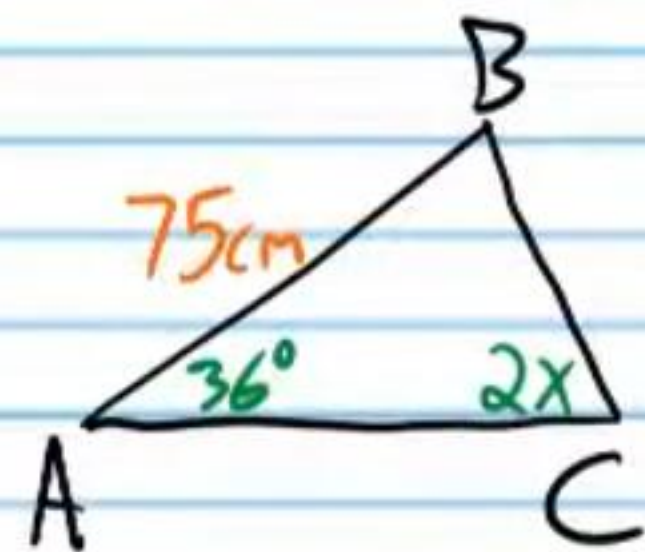
7, 4

⑪ In the problems below, determine if the triangles are congruent. If they are congruent, write a proper congruence statement and the triangle postulate or theorem that proves that they are congruent. You do not have to write a formal proof (i.e. a two-column proof).

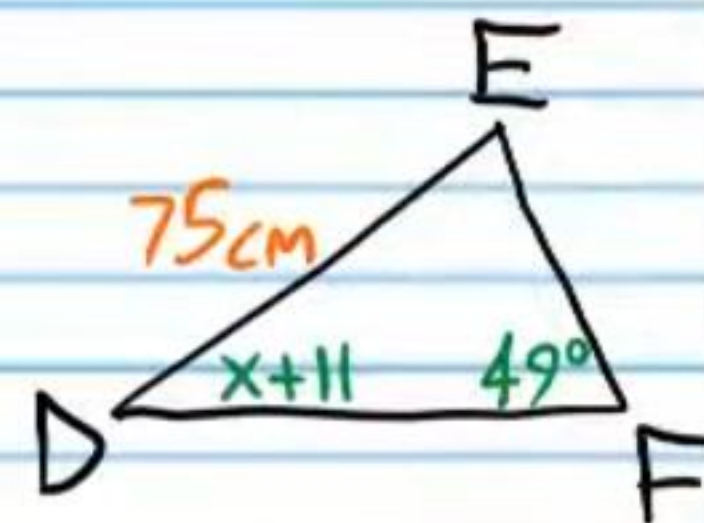
i) ABCDEF is a regular hexagon. ii) $x = 25$.



$\triangle ABC \cong$?

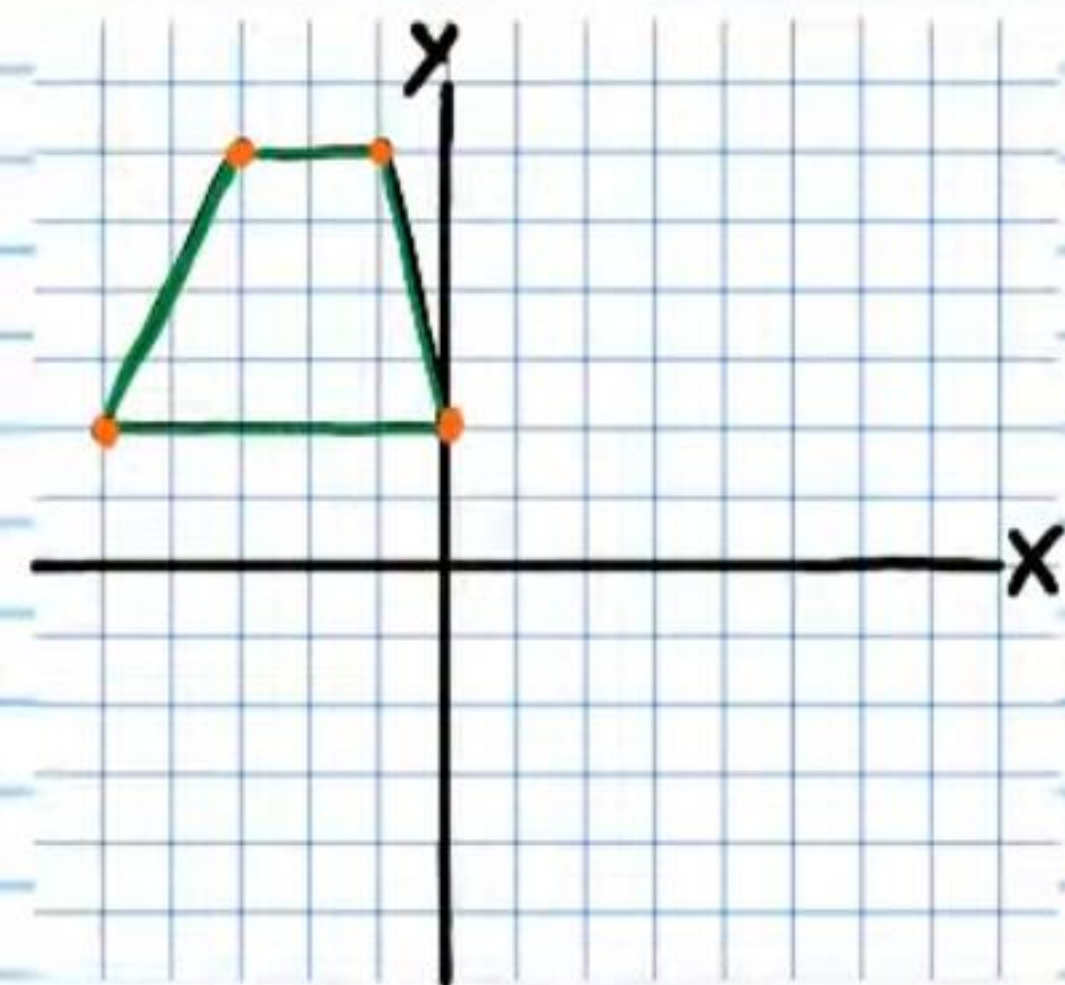


$\triangle ABC \cong$?

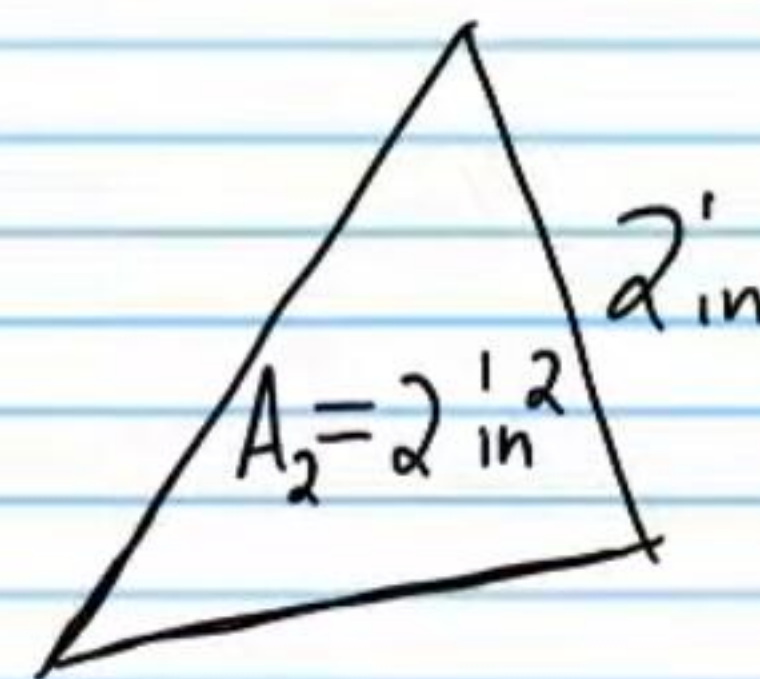
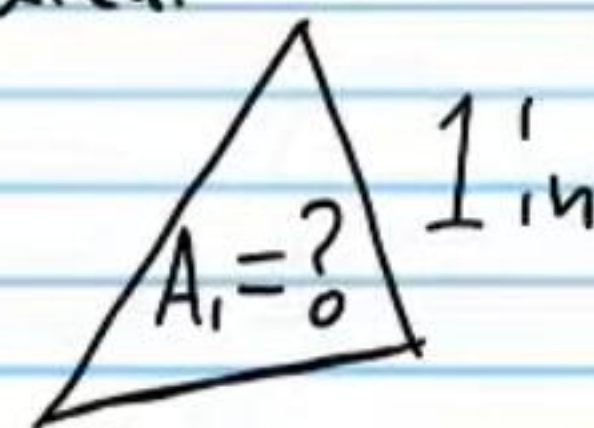


⑫ Given the pre-image, graph the image.

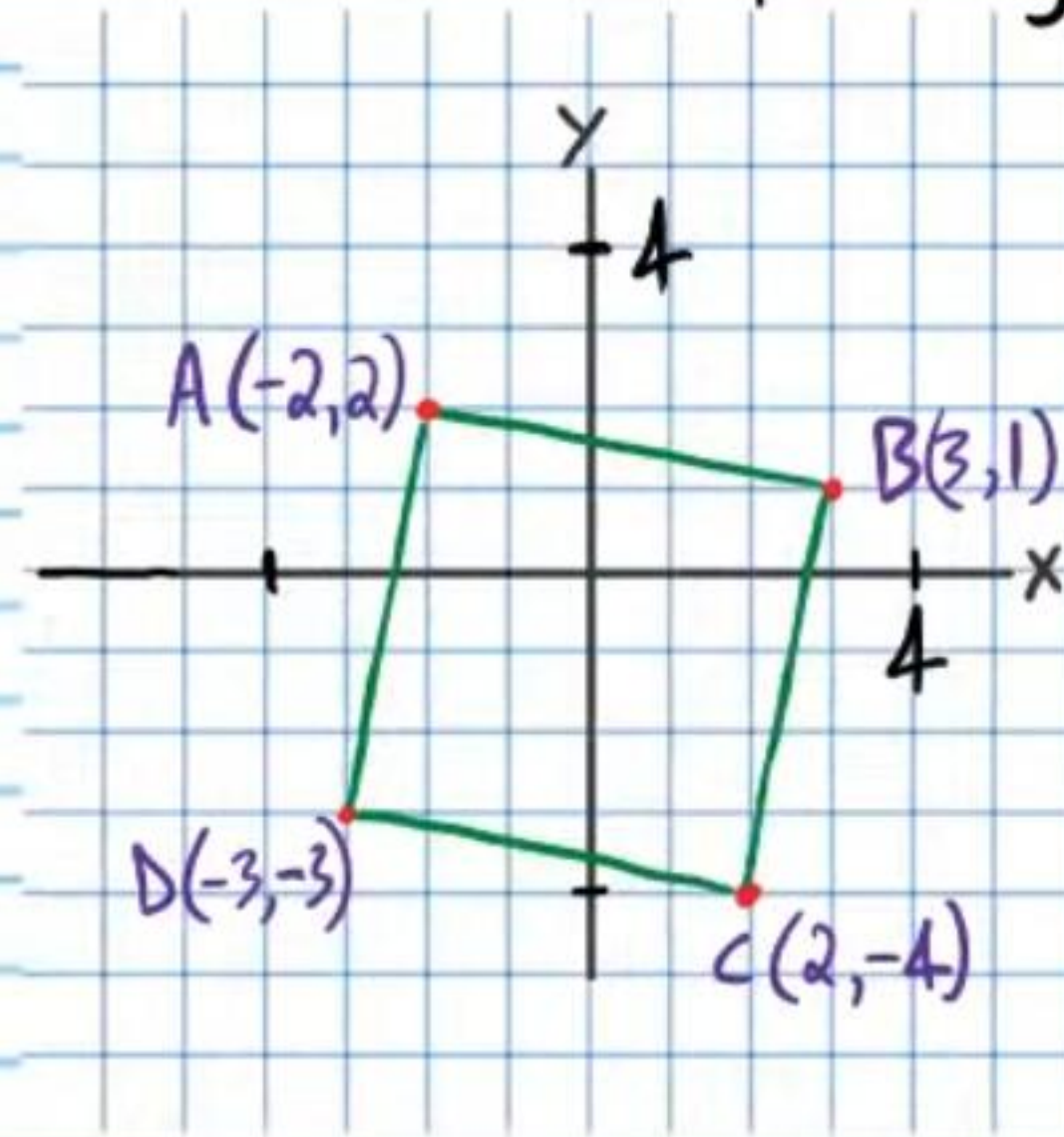
$$(x, y) \rightarrow (x+3, y-1)$$



⑬ If the polygons below are similar, find the missing area.

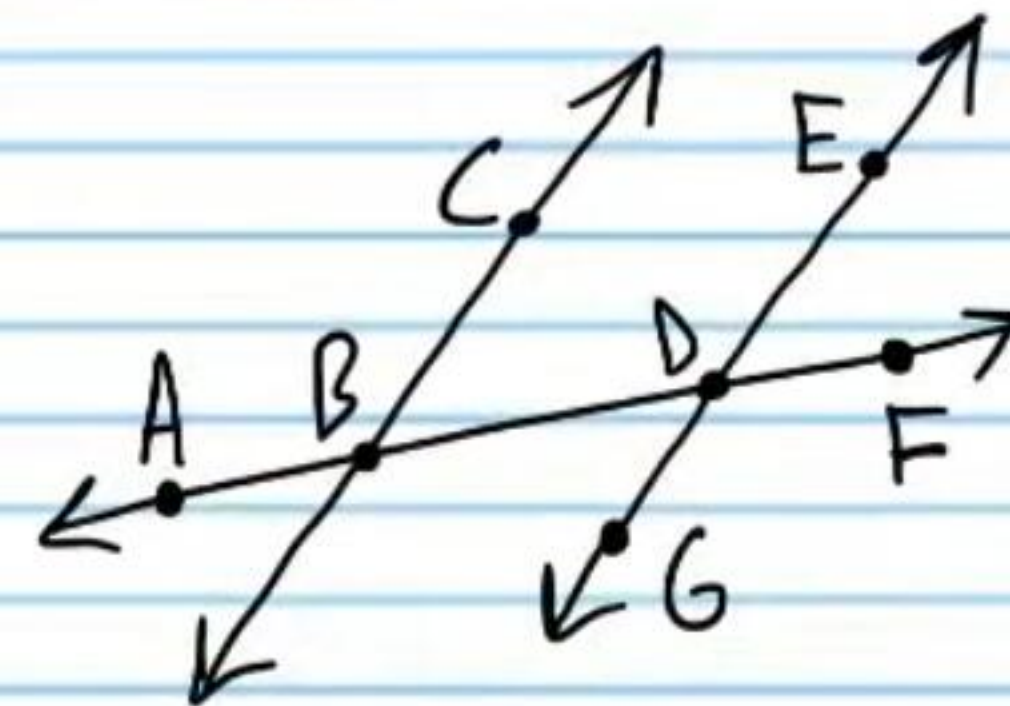


②0 Show that the quadrilateral below is a parallelogram.



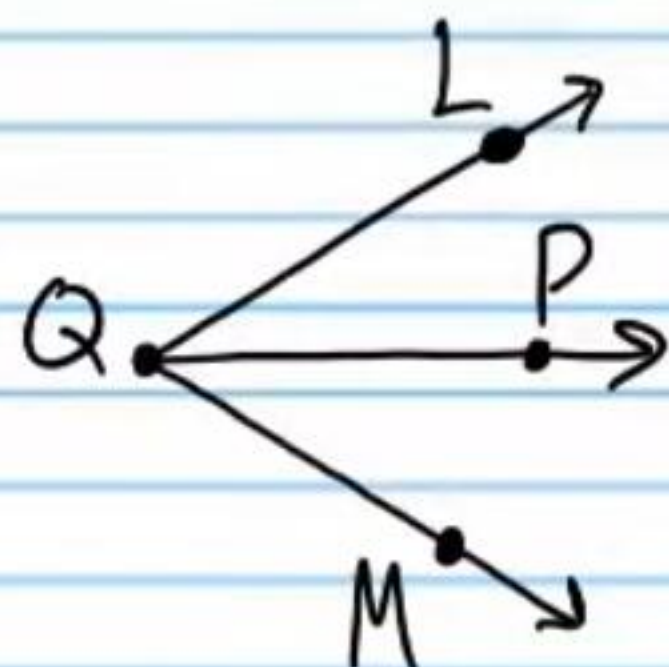
Show that the following statements are true using two-column proofs. You might not use the same number of steps written in Roman numerals. You may use more or less.

②1 If $\overleftrightarrow{BC} \parallel \overleftrightarrow{DE}$ below, use the Corresponding Angles Postulate and the Vertical Angles Theorem to prove the Alternate Exterior Angles Theorem (i.e. that $\angle ABC \cong \angle GDF$).



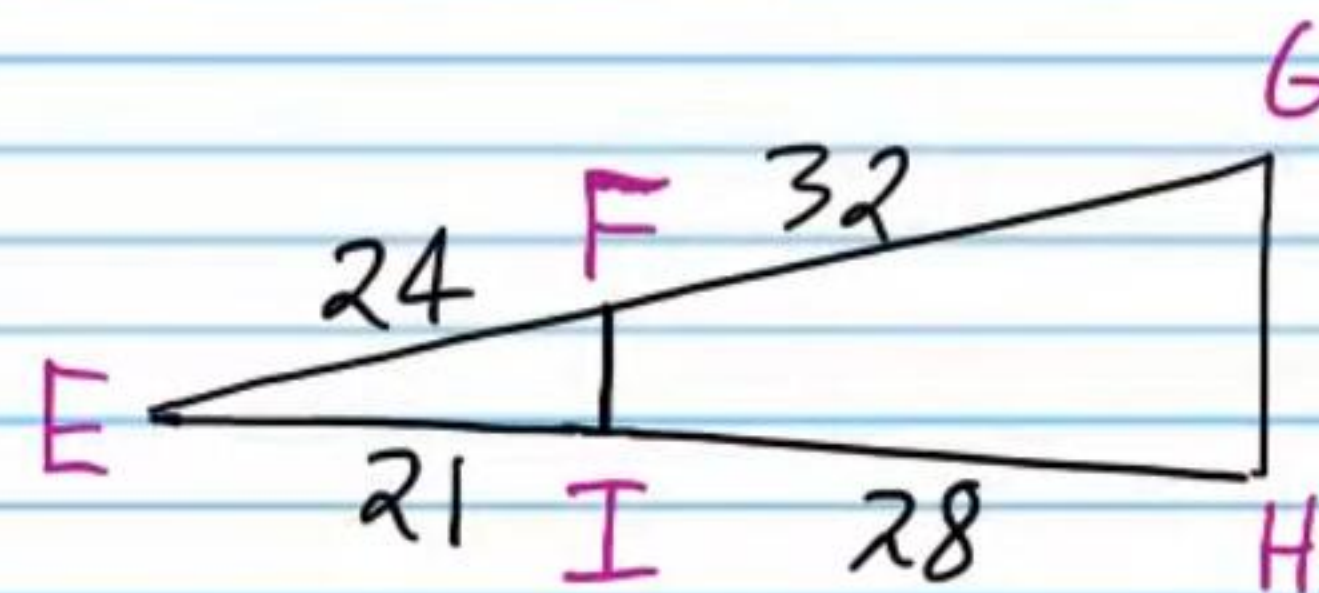
Statement	Reason
I)	
II)	
III)	
IV)	

②② If $\overrightarrow{QP}$ bisects $\angle LQM$, then $m\angle LQP = \frac{1}{2}m\angle LQM$.



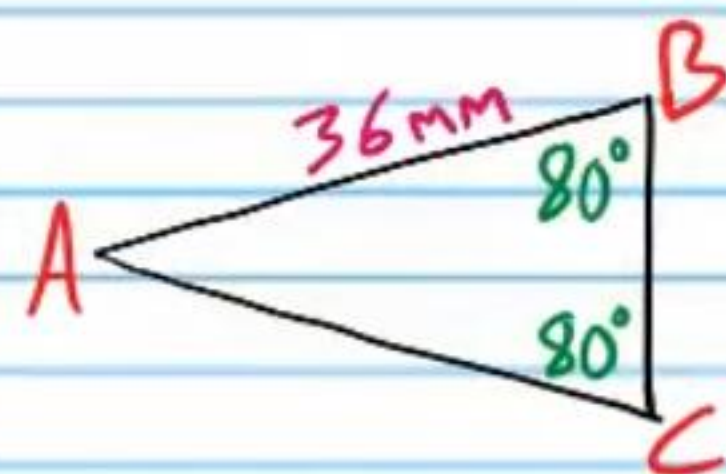
Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	
VII)	

②③ $\triangle EFI$ and $\triangle EGH$ are similar.



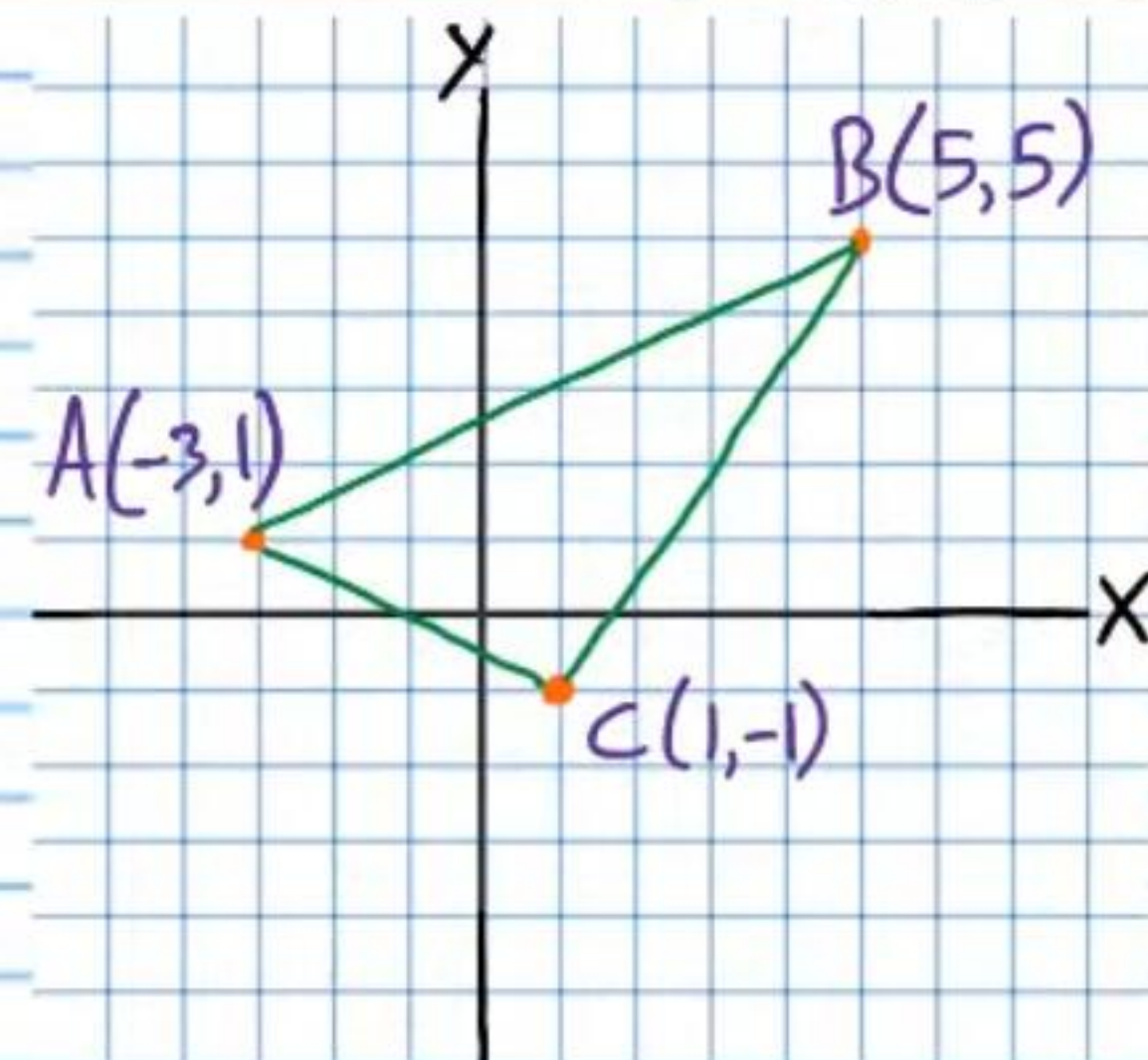
Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	
VII)	
VIII)	
IX)	

24) If $m\angle B = 80^\circ$ and $m\angle C = 80^\circ$ in the triangle below, then $AC = 36\text{ mm}$. *Hint: You must use the Isosceles Triangle Converse Theorem.*



Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	
VII)	

25) Find the coordinates of the circumcenter of the triangle below.



Geometry Final Test (Part I) Practice Version 2 Answers

① $x = 20m$

② i) $x = \frac{3}{5}$

ii) Not Enough Information

③ 30,000

④ (8, 15)

⑤ Walter will give Alice a gift for Valentine's Day this year.

⑥ i) The Law of Syllogism

ii) Deductive

⑦ Possible Answers:

- Fido may have jumped in a pond.
- He ran through a river.
- Someone squirted him with a hose.

⑧ i) a

ii) b

iii) c

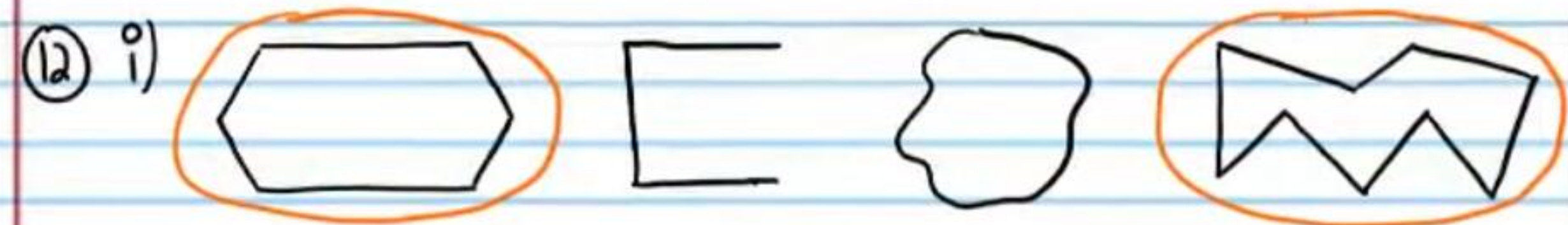
iv) ABC or m.

Letter order
does not
matter.

⑨ If it is raining, then I will not stay home.

⑩ $x = 4, y = 90^\circ$

⑪ DE, EF, DF



ii) No. Although the angles share a vertex, they do not share a side.

⑬ i) Equilateral

ii) Equiangular or acute

⑭ a) $m\angle MBC = 52^\circ$

b) $MJ = 5ft$

c) $m\angle C = 30.76^\circ$

d) $AM = 13ft$

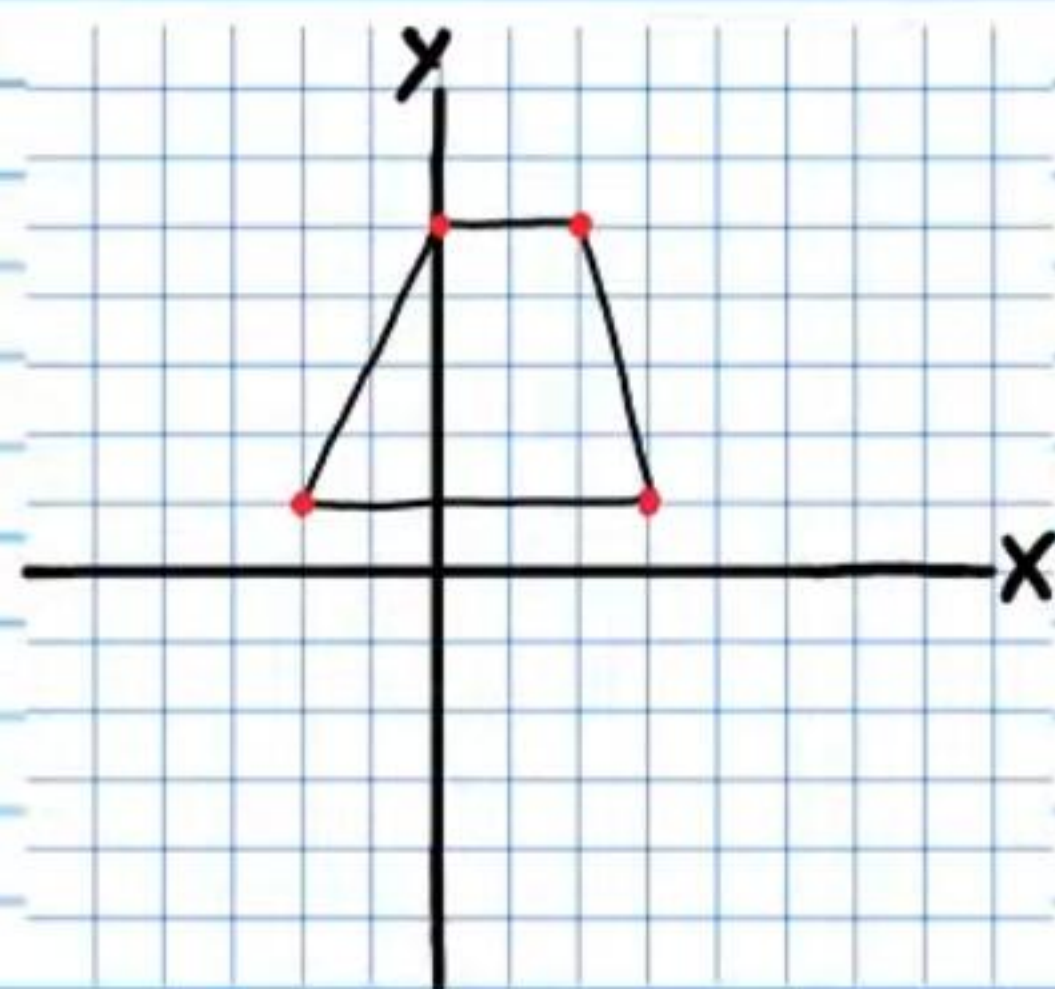
⑮ $x = 30$

⑯ $3 < x < 11$

⑰ i) $\triangle ABC \cong \triangle CDE$ or $\triangle ABC \cong \triangle EDC$, SAS

ii) Not congruent

(18)



(19) $A_1 = \frac{1}{2} \text{ in}^2$

(20) $m_{AB} = \frac{2-1}{-2-3} = -\frac{1}{5}$

$m_{DC} = \frac{-3-(-4)}{-3-2} = -\frac{1}{5}$

$\overline{AB} \parallel \overline{DC}$

$m_{BC} = \frac{1-(-4)}{3-2} = \frac{5}{1} = 5$

$m_{AD} = \frac{-3-2}{-3-(-2)} = \frac{-5}{-1} = 5$

$\overline{BC} \parallel \overline{AD}$

Therefore, the quadrilateral is a parallelogram.

(21)

Statement	Reason
I) $\vec{BC} \parallel \vec{DE}$	Given
II) $\angle ABC \cong \angle BDE$	CAP
III) $\angle BDE \cong \angle GDF$	Vert. $\angle$'s Thm
IV) $\angle ABC \cong \angle GDF$	Trans. Prop of $\cong$

(22)

Statement	Reason
I) $\vec{QP}$ bisects $\angle LQM$	Given.
II) $m\angle LQM = m\angle LQP + m\angle PQM$	$\angle$ Add. Post.
III) $m\angle LQP = m\angle PQM$	Def. of $\angle$ bisector.
IV) $m\angle LQM = m\angle LQP + m\angle LQP$	Subs. Prop of $=$.
V) $m\angle LQM = 2m\angle LQP$	Simplification.
VI) $\frac{m\angle LQM}{2} = m\angle LQP$	Div. Prop of $=$.
VII) $m\angle LQP = \frac{m\angle LQM}{2}$	Sym. Prop of $=$.

Statement	Reason
I) $EF = 24, FG = 32, EI = 21, \& IH = 28.$	Given (diagram)
II) $EG = EF + FG \& EH = EI + IH.$	Seg. Add. Post.
III) $EG = 24 + 32 \& EH = 21 + 28.$	Subs. Prop of =.
IV) $EG = 56 \& EH = 49$	Simplification.
V) $\frac{EF}{EG} = \frac{24}{56} \& \frac{EI}{EH} = \frac{21}{49}$	Subs. Prop of =.
VI) $\frac{EF}{EG} = \frac{3}{7} \& \frac{EI}{EH} = \frac{3}{7}$	Simplification.
VII) $\frac{EF}{EG} = \frac{EI}{EH}$	Trans. Prop of =.
VIII) $\angle E \cong \angle E$	Ref. Prop of $\cong$.
IX) $\triangle EFI \sim \triangle EGH$	SAS

Statement	Reason
I) $m\angle B = 80^\circ \& m\angle C = 80^\circ$	Given (diagram)
II) $m\angle B = m\angle C$	Trans. Prop of =.
III) $\angle B \cong \angle C$	Def. of $\cong \angle$'s
IV) $\overline{AC} \cong \overline{AB}$	Isos. $\triangle$ Converse Thm.
V) $AC = AB$	Def. of $\cong$ segs.
VI) $AB = 36 \text{ mm}$	Given (diagram)
VII) $AC = 36 \text{ mm}$	Trans. Prop of =.

(25) $\left(\frac{3}{4}, 3\frac{1}{2}\right)$ or $(0.75, 3.5)$